Module 3F Questions:

1. Researchers want to examine whether **activity level** is associated with **PPARG expression** in a small sample of mice. Both variables are measured on ordinal or continuous scales, but the relationship may not be linear, so they decide to use **Spearman's rank correlation**.

   They collect the following data:

Mouse	Activity Score	PPARG Expression
M1	3	6
M2	3.5	6
M3	6.0	8
M4	6.5	8
M5	4.0	6

Conduct a Spearman's Rank hypothesis, but remember to do the following steps (besides Ho/Ha and the other hypothesis testing steps)

1. **Rank the activity scores** from lowest to highest.
2. **Rank the PPARG levels** from lowest to highest.
3. Determine whether the relationship between activity and PPARG is:
   - positive
   - negative
   - or zero
4. When would you choose Spearman over Pearson?